

Lecture XII - Metro Passenger Flow Control

Applied Optimization with Julia

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Introduction

FIFA World Cup 2022 in Qatar

Learning Objectives

After this lecture, you will be able to:

- **Model** time-expanded passenger flows with precomputed delays
- **Translate** a safety goal into inflow control constraints
- **Explain** how the set $\mathcal{R}_{e,t}$ connects periods and minutes
- **Embed** an optimization model into a rolling-horizon heuristic

Public transport

- FIFA World Cup 2022 took place in a small region
- The capacity of the metro system was **finite**
- More than 1 million tourists were expected
- Metro usage was **free for all ticket holders**
- Transport methods were expected to be **overloaded**

...

Question: **What could become an issue?**

Crowd Disasters

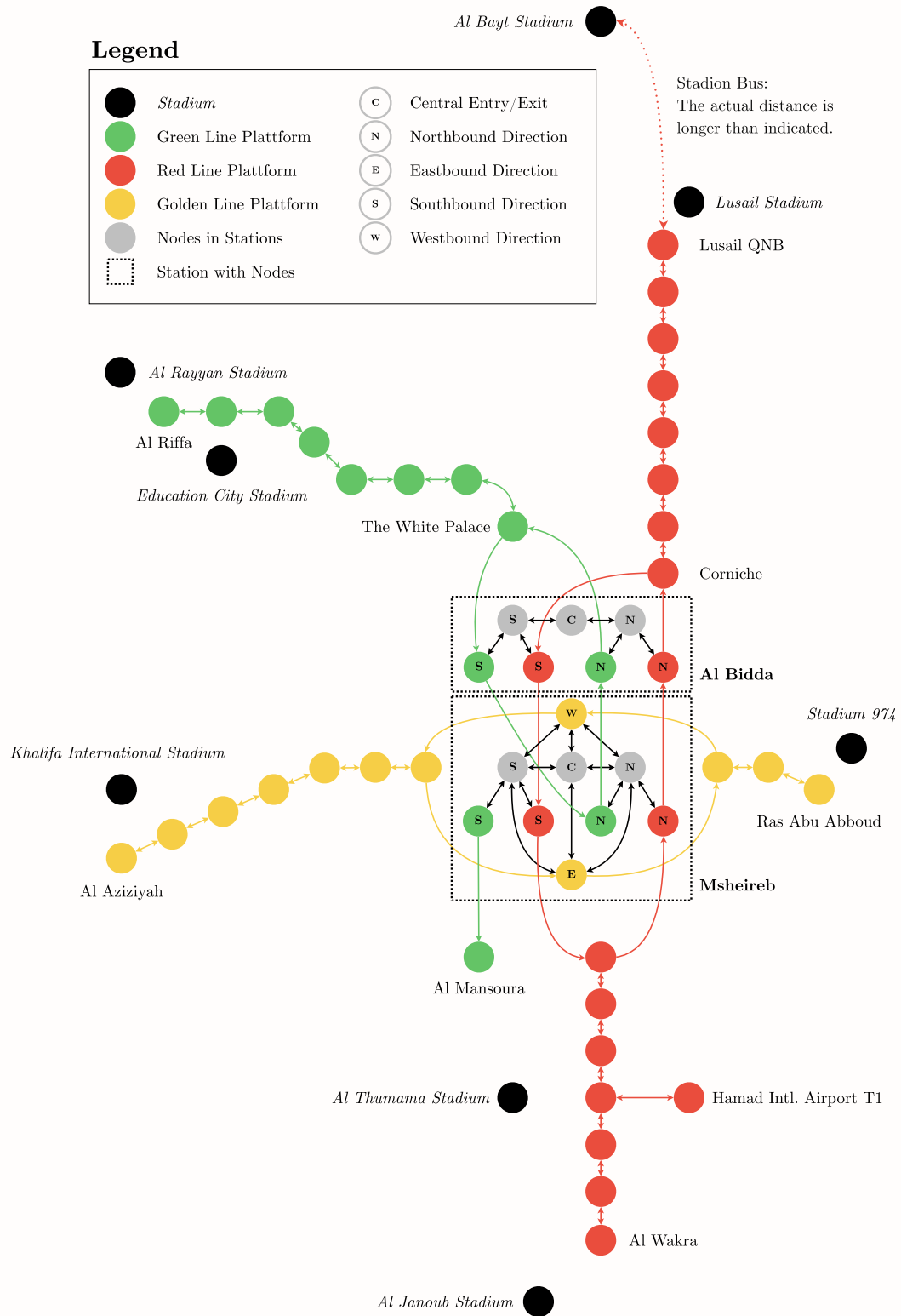
Potential locations

Question: **Where could crowd disasters happen?**

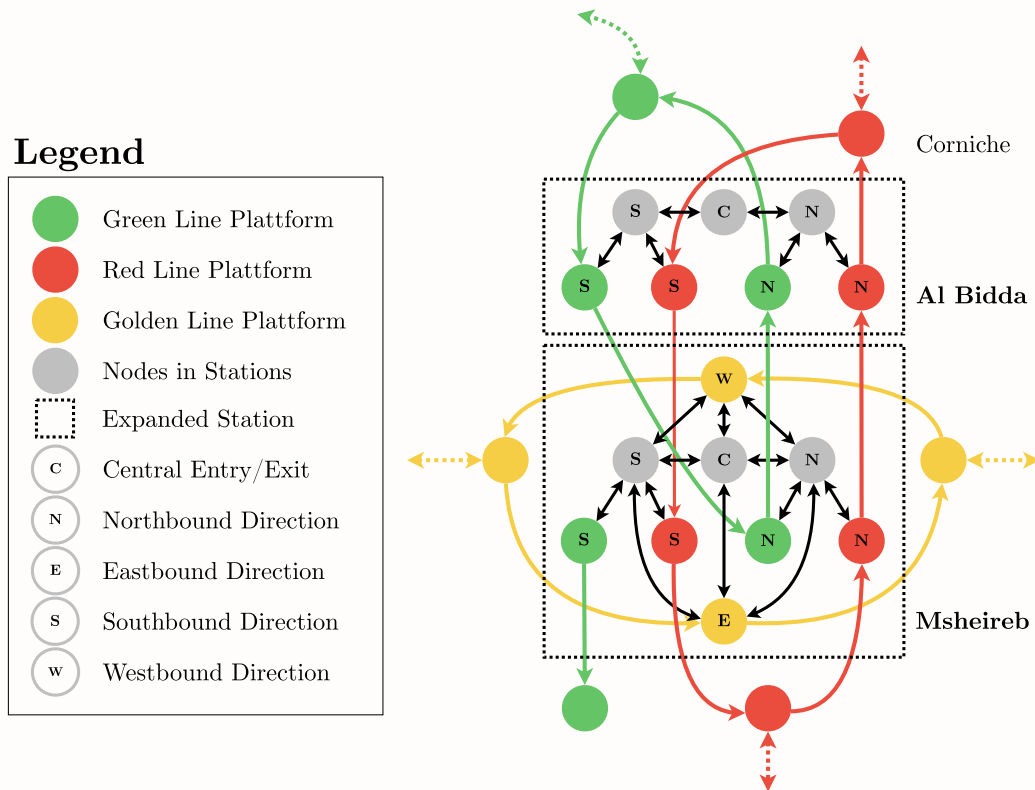
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- At event venues and in waiting areas
 - At intersections of multiple pedestrian flows
 - In narrow passages and entrances
 - On crowded metro platforms and transfer stations
 - At ticket/turnstile bottlenecks and emergency exits
-

Metrosystem of Doha



Closer look at the Metro System



Main Bottleneck

Question: **What could be a potential bottleneck?**

...

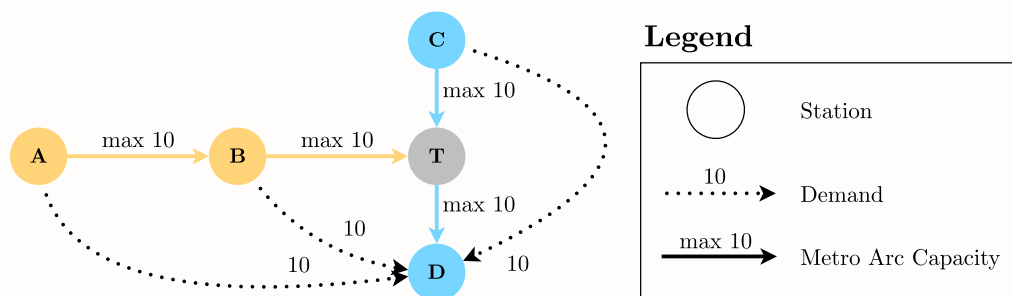
- Red Line has twice the capacity of Green and Gold Line
- People use the metro to get to the event venues

...

Question: **What could be the issue?**

- Imagine a match at a stadium along the Gold line
- People from Red and Green line will try to get there
- This will cause a massive crowd at transfer stations

Visualization



- 10 people want to start per minute at **each station**
- Everybody wants to get to **station D**

Capacities inside the stations

- Capacities **inside the stations** are limited as well
- These include the escalators, stairs, and elevators
- But also the platforms and the ticket gates

...

⚠ Warning

This can lead to overcrowding and potential crowd disasters!

General Issues

Question: **What could also be an issue?**

...

- Often unknown **how many people** exactly will participate
- Gathering data is extremely important and difficult
- Crowd behavior can be unpredictable and dynamic
- Weather conditions may affect **transportation preferences**
- Cultural factors influence **crowd movement patterns**
- Emergency situations require **flexible contingency plans**

Let's take a

closer look at the

problem structure!

Problem Structure

Objective

Question: **What could be the objectives of the authorities?**

...

- A safe and successful event as host country
- Good publicity and a positive recognition worldwide
- Satisfied visitors that enjoyed their time

...

 Note: We can model none of the above directly!

But we can assist in the estimation of transport demands and the design of operational plans to ensure public **safety** and a **smooth transportation** through the city.

Underlying Problem

Question: **Can we just start modeling?**

...

- First, we need to understand the **movement patterns**
- Event is **unprecedented**, movement patterns are unknown
- Multiple concurrent events affect flow patterns

...

 Tip: We need to estimate the data first!

This is a **huge challenge**, but we can use **simulation** to estimate the data.

Simulation

- Based on publicly available data from the area
- Simulates **all individuals** participating at the event
- Includes **all transportation infrastructure**
- Individual mode choice based on a choice model

...

 Tip: Julia is a great tool for this!

The simulation was built in Julia. With 1,000,000 individuals walking, using cars, buses, and the metro, simulating a day took **less than 5 minutes**.

Results of the Simulation



- Detailed movement patterns throughout the region
- Potential sections at risk in the transport infrastructure
- Potential capacity overloads at event locations

But it is still
based on a lot
of assumptions!

Main risk: Metro

Idea

Question: **What can we do to prevent crowd disasters?**

...

- Regulate the **inflow** at each individual station
- Ensure utilized capacity is always within bounds

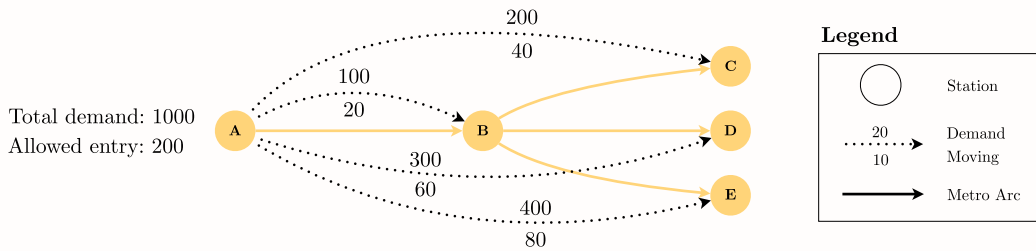
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Question: **What could we try to model?**

...

- Minimize the queues outside of the metro stations
- Based on the **allowed inflow** and origin-destination data
- Adhering to the **capacity constraints**

Difficulty



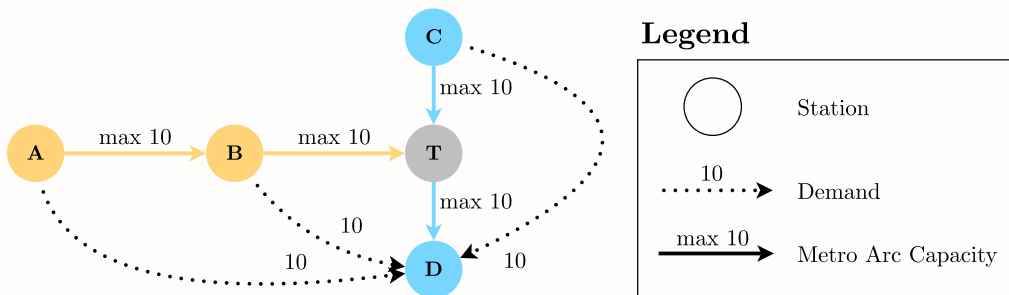
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- Patterns have different origin-destination pairs
- Regulating the inflow does not affect the destination

Model Formulation

Graph Sets?

Question: **What could be the sets for the graph?**



Graph Sets?

- $\mathcal{G}$ - Connected digraph of the metro network $\mathcal{G} = (\mathcal{O}, \mathcal{E})$
- $\mathcal{O}$ - Set of metro stations, indexed by o
- $\mathcal{E}$ - Set of directed arcs between connected stations

...

i Note: These are the easier sets.

These simply help us to represent the entire metro system as a graph with different nodes $\mathcal{O}$ and arcs $\mathcal{E}$ between connected stations.

Time Sets?

Question: **What could be the time sets?**

- $\mathcal{T}$ - Set of minutes in the time horizon, indexed by t
- $\mathcal{P}$ - Set of periods in the observed time horizon, where $p \in \{1, 2, \dots, n\}$

...

i Note: Number of periods

We further define n as the **number of periods in observed time horizon**.

Periods

Question: **Why do we add periods here?**

...

- Staff needs **clear**, consistent instructions
- Frequent changes in flow increase risk of errors
- **Easier** to manage and communicate for planners

...

i Note: Period Length

We define the period length as the length of the period in minutes measured as m minutes, which is the **same for all periods**.

Mapping Minutes to Periods

- We can define an **additional set** with their relation
- It specifies the relation of periods p to minutes t .

$$I_p = \{t \in \mathcal{T} \mid (p-1) \times m + 1 \leq t \leq p \times m\} \quad \forall p \in \mathcal{P}$$

...

Question: **Can anybody explain it?**

...

- I_p is the set of minutes t that belong to period p
- Minutes are **not overlapping**, but they are **continuous**

Parameters?

Question: **What could be possible parameters?**

...

- $q_{o,d,p}$ - Queued passengers from station o to d with $o, d \in \mathcal{O}$ in p
- τ_e - Travel time (min) of the arcs $e \in \mathcal{E}$
- c_e - Max. allowed arc entry rate e per minute with $e \in \mathcal{E}$
- c_o^{min} - Min. station entry rate o per minute with $o \in \mathcal{O}$
- c_o^{max} - Max. station entry rate o per minute with $o \in \mathcal{O}$
- α - Safety factor applied to all capacities ($0 < \alpha < 1$)

Metro Movement

Question: **How do people move inside the metro?**

...

- Simple network: assume people use the **shortest path**
- Use Dijkstra's algorithm to compute it
- From each station to all other stations

...

💡 Tip: We can save the results in a new set $\mathcal{C}_{o,d}$

This will help us to model the movement inside the metro.

Shortest Paths (SP)

- $\mathcal{C}_{o,d}$ - Set of arcs $e \in \mathcal{E}$ on the SP from o to d with $o, d \in \mathcal{O}$

...

i Note

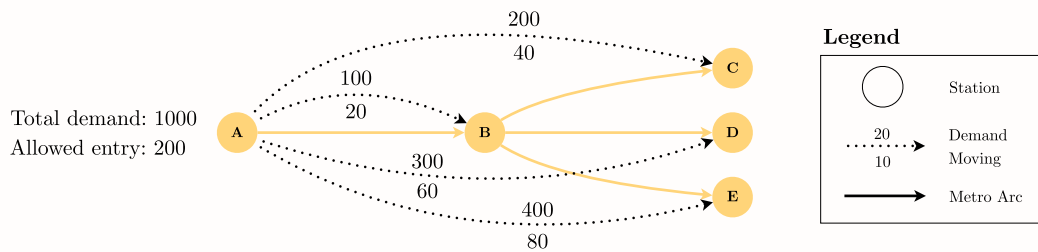
The arc travel times τ_e are only the **input** to Dijkstra's algorithm. From the resulting paths we compute the travel time parameter the model actually uses:

...

- $\tau_{o,e}$ - Travel time (min) on the SP from station $o \in \mathcal{O}$ to arc $e \in \mathcal{E}$

People Spreading

Question: **How do people spread?**



Ratio of Origin-Destination

- We **cannot control** the destination of the passengers
- Thus we assume that people spread based on destination

...

$$\frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \quad \forall o, d \in \mathcal{O}, p \in \mathcal{P}$$

...

i Note

Based on the ratio of the different destinations d to the total queue for each station $o \in \mathcal{O}$ in each period $p \in \mathcal{P}$. Note that the summation needs its **own index** f over all destinations, as d is already taken by the ratio's destination.

Variables and Objective

Decision Variable?

! Our goal is to:

Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints**.

💡 Tip

There is **one queue for all passenger flow directions**, as managing multiple queues would be too complex for the planners!

Decision Variable

i We need the following sets:

- All the metro stations, $o \in \mathcal{O}$
- All periods under observation, $p \in \mathcal{P}$

Question: **What could be our decision variable?**

...

- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p

Objective Function?

! Our main objective is to:

Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints**.

...

Question: **How again are queues minimized?**

...

- By the allowed inflow $X_{o,p}$ subtracted from the queue

Objective Function

i We need the following parameters and variables:

- $q_{o,d,p}$ - People queued to travel from station o to d with $o, d \in \mathcal{O}$ in period p
- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p

...

Question: **What could be our objective function?**

...

$$\text{Minimize } \sum_{o \in \mathcal{O}} \sum_{p \in \mathcal{P}} \left(\sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p} \right)$$

Objective Function Revisited

Question: **What does the solver really maximize here?**

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- The queued passengers $q_{o,d,p}$ are parameters, not variables
- Their sum is a **constant** — the solver cannot change it

...

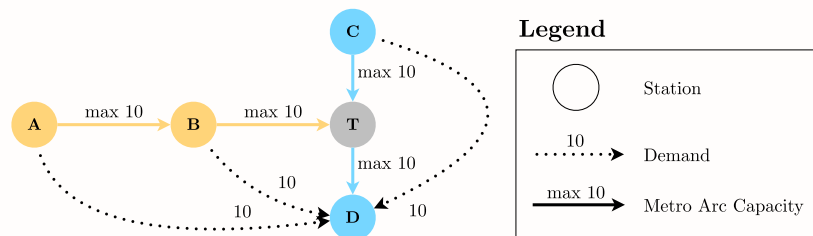
i Note: Minimizing queues = maximizing admissions

Minimizing the objective is exactly the same as **maximizing the total admitted passengers** $\sum_{o \in \mathcal{O}} \sum_{p \in \mathcal{P}} m \times X_{o,p}$. This is also why the model gets away without explicit queue dynamics.

Constraints

Necessary Constraints

Question: **What constraints do we need?**



...

- The capacity of each arc is **not exceeded**
- Do not dispatch more people than are queued
- Do not dispatch less than the minimum allowed inflow

Things are going to
get a little
complicated now!

Central Question

Question: **When do people flowing into the metro system change the arcs?**

...

- People enter metro station at o with destination d
- They will lead to a usage of an arc e on their SP
- This usage depends on their path and the travel times

...

💡 Tip: You'll likely know what we need now!

We can add a new set $\mathcal{R}_{e,t}$ to help us with this.

Set of Time-Delays

$$\mathcal{R}_{e,t} = \{(o, d, p) \mid o, d \in \mathcal{O}, p \in \mathcal{P}, q_{o,d,p} > 0, \\ e \in \mathcal{C}_{o,d}, t - \tau_{o,e} \in I_p\} \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

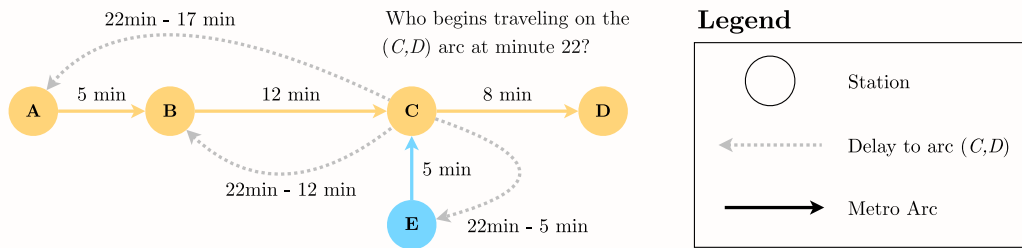
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Question: **Who can explain this set?**

...

The set $\mathcal{R}_{e,t}$ contains **all combinations** (o, d, p) which trigger a **capacity utilization** of arc e in minute t .

Small Example



...

💡 Tip

The set contains all o-d pairs and periods that result in passengers entering arc (C, D) at minute 22. With $m = 2$, it is $\{(A, D, 3), (B, D, 5), (C, D, 11), (E, D, 9)\}$.

Check it yourself, e.g. from A : $\tau_{A,(C,D)} = 5 + 12 = 17$ and $22 - 17 = 5 \in I_3 = \{5, 6\}$.

Essentially, we can use this set to compute the utilization of an arc at each minute based on the inflow level at the different stations and periods.

Ensure Capacity Utilization?

! The goal of this constraint is to:

Ensure that the capacity of each arc is not exceeded at any minute.

...

i We need the following variable, parameter and set:

- $q_{o,d,p}$ - people queued to travel from station $o \in \mathcal{O}$ to station $d \in \mathcal{O}$ in p
- c_e - people max. allowed to enter arc e per minute with $e \in \mathcal{E}$
- $\mathcal{R}_{e,t}$ - mapping of station entries to arc e in minute t with $e \in \mathcal{E}$ and $t \in \mathcal{T}$
- α - safety factor applied to all capacities ($0 < \alpha < 1$)
- $X_{o,p}$ - the allowed inflow per minute at metro station o in the period p

Ensure Capacity Utilization

Question: What could be the constraint?

...

$$\sum_{(o,d,p) \in \mathcal{R}_{e,t}} X_{o,p} \times \frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \leq \alpha \times c_e \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

...

i Note

Here, we combine the inflow at each station based on the o-d proportion and let the people spread through the network, checking that no arc is over-utilized at any minute.

Bound the Inflow Rate?

! The goal of this constraint is to:

Ensure that the inflow rate at each station stays within operational bounds.

...

i We need the following variables and parameters:

- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p
- c_o^{min} - Min. station entry rate o per minute
- c_o^{max} - Max. station entry rate o per minute
- α - Safety factor applied to all capacities ($0 < \alpha < 1$)

Bound the Inflow Rate

Question: **What could be the constraint?**

...

$$c_o^{min} \leq X_{o,p} \leq \alpha \times c_o^{max} \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

...

Note

This constraint bounds the inflow rate at each station between the **minimum and maximum** allowed entry rates, ensuring compliance with **operational capacity limits**.

Dispatch Only Available People?

 The goal of this constraint is to:

Ensure that we do not dispatch more people than are queued.

...

We need the following variables and parameters:

- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p
- $q_{o,d,p}$ - People queued to travel from station o to d with $o, d \in \mathcal{O}$ in p
- m - period length in minutes

Dispatch Only Available People

Question: **What could be the constraint?**

...

$$\sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p} \geq 0 \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

...

Note

This constraint ensures **non-negative remaining queues**, preventing the solver from dispatching more passengers than available in the queue.

A Hidden Pitfall

Question: **What if a queue is smaller than $m \times c_o^{min}$?**

...

- The lower bound forces $m \times X_{o,p} \geq m \times c_o^{min}$
- But we may only dispatch up to $\sum_{d \in \mathcal{O}} q_{o,d,p}$ people
- Example: $c_o^{min} = 2$, $m = 15$, but only 20 people queued
- A quiet station late at night makes the model infeasible!

...

Question: **How could we fix this?**

...

- Set $c_o^{min} = 0$, cap the bound at the queue, or soften the constraint

Metro Inflow Model

subject to:

Model Characteristics

Characteristics

Questions: **On model characteristics**

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?

...

i Note: The answers

It is a linear program, as objective and constraints are linear. The variable $X_{o,p} \geq 0$ is **continuous** — fractional inflow rates per minute are perfectly acceptable here.

Model Assumptions

Questions: **On model assumptions**

- What assumptions have we made?
- What are likely issues that can arise if applied?
- Have we thought in detail about queues?
- Are shortest paths a feasible assumption?

Implementation and Impact

Metro Inflow Optimization

...

Question: **Can this be applied?**

Metro Inflow Problem

- Solved very fast within seconds for realistic problem sizes
- **But** we cannot plan or control the metro inflow

- Queues are **too simplified** with passengers disappearing

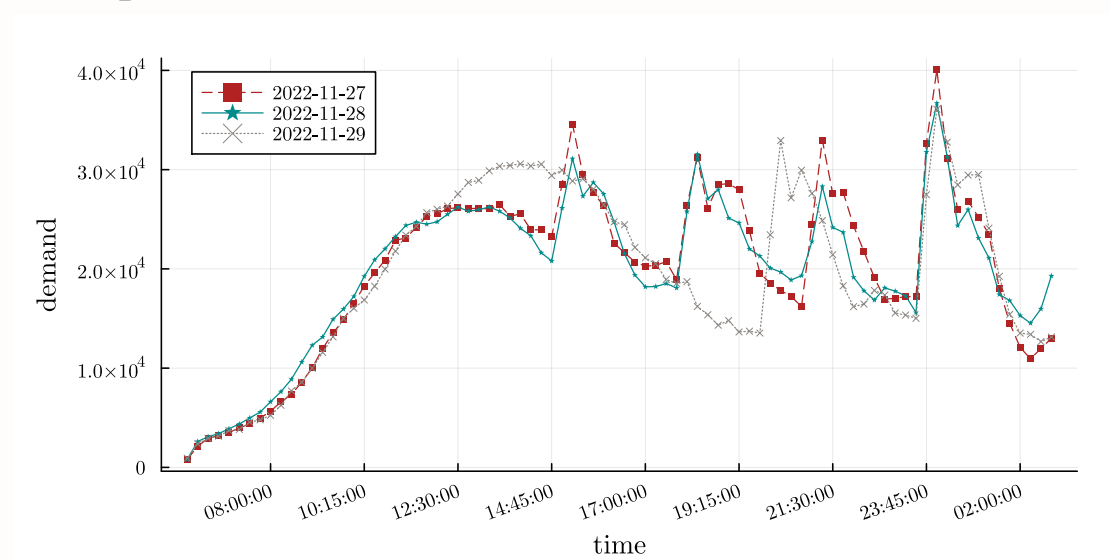
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Question: **Any ideas, how the current model could be improved or how it could be embedded into a heuristic?**

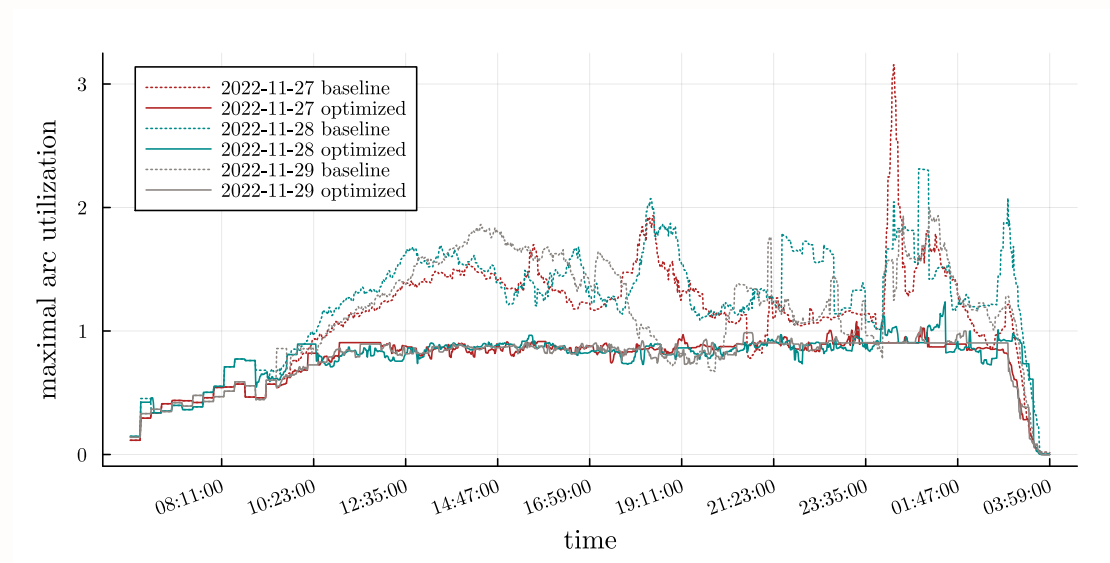
Heuristic: Step-Wise

- Solve the model for the time-horizon of a **few periods**
- Fix the inflow in the **current first period**
- Decrease capacity in the network **based on the inflow**
- Transfer **remaining queues** into the subsequent period
- Solve the model again
- **Repeat**, until the inflow is computed for all periods

Transport Demand



Utilization Analysis



Key Results

- Model reduces capacity violations from 25,941 to just 4
- Maximum arc utilization drops from **316% to 109%**
- 81% of passengers accommodated safely
- Computation time: **~1.4 seconds** per planning period

...

i Note: Safety vs. Throughput Tradeoff

The remaining 19% of passengers experience external queuing - this is the price for preventing dangerous overcrowding inside the system.

Didn't We Bound the Utilization?

Question: **Our constraint enforces $\alpha \times c_e$ with $\alpha < 1$. How can utilization still reach 109%?**

...

- The constraint binds the model for the forecast demand
- The 109% occurs in the **rolling-horizon simulation** with realized demand
- Realized flows can deviate from the forecast — but far less than **316%**

Scalability: Shanghai Metro

- Validated on **302 stations** (~8× more than Doha's 37)
- 99% reduction in capacity violations
- Computation: **< 2 minutes** per period
- Works for routine **peak-hour demand**

...

💡 Tip: Generalizable Approach

The method scales from event-driven demand (Qatar World Cup) to routine metropolitan operations (Shanghai rush hour).

Implementation

- **Assumption of known destinations** is strong
- Movements **seemed to follow our forecasts**
- We did achieve our goal of **metro inflow control**
- Simulation was used to **estimate the inflows**

...

i Note

Few dangerous situations, especially at the FIFA Fan Fest, were **handled well** by the authorities.

Comparison with Others

Feature	Other Methods	Our Approach
FIFO Enforcement	Rarely	Yes
Single Queue	Often separate	Yes
Real-Time Capable	Some	Yes (seconds to minutes)
Network Scale	Up to 327 stations	Comparable (302 stations)

...

i Note: State-of-the-Art

First network-level approach combining FIFO enforcement, single-queue modeling, and real-time computational performance.

Wrap Up

- Model can help to achieve a good balance
- Can be adapted easily to **any metro system worldwide**
- Especially interesting for **larger Asian cities**

...

i And that's it for today's lecture!

We now have covered a metro inflow control problem based on a real-world application and are ready to start solving some new tasks in the upcoming tutorial.

Questions?

Literature

Literature I

For more interesting literature to learn more about Julia, take a look at the [literature list](#) of this course.

Bibliography